

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	ME-ENVE
Program Name	Master of Engineering in Environmental Engineering
Assessment Year	2024-2025
Program Assessment Coordinator	Brent Stephens
Year of Current Assessment Plan	2024-2025
No Assessment This Year per Plan	☐
LO(s) Assessed	3
Presented at program authority meeting date: <i>Please attach minutes</i>	n/a

Methodology

The Master of Engineering in Environmental Engineering program is a coursework-based graduate degree program oriented toward students who wish to develop more knowledge in multiple areas of environmental engineering. The program provides students with opportunities to gain experience in a variety of environmental engineering domains, including water and wastewater treatment, water resources engineering, indoor and outdoor air quality, hazardous waste management, environmental health and safety and industrial hygiene, and environmental sustainability.

The program is intended for preparation for advanced engineering practice rooted in the principles of physics, chemistry, biology, and mathematics to develop and analyze solutions to environmental problems, including air and water pollution control, water and wastewater treatment systems design, climate change, resource depletion, and more.

The primary learning objectives for this program are for students to:

1. Understand contaminant fate and transport in the environment.
2. Formulate mass and energy balances and apply modeling concepts to practical environmental science and engineering problems.
3. Assess the sustainability of engineered systems.

Artifact(s) used to evaluate student learning: Performance on assignments and exams in one of the required courses in the program (typically ENVE 506 Chemodynamics, ENVE 522 Global Environmental Change and Sustainability Analysis, or ENVE 542 Physicochemical Processes in Environmental Engineering)

Semester(s) in which artifacts are collected and assessed: End of each academic year

Sample size: Each student in the program who took the selected course (ENVE 506, 522, and 542 are typically on rotation and will thus alternate in offering and assessment schedule)

Data to report: Results are reported as statistical distributions (i.e., histograms) and summary statistics for the sample of students in each year on each performance indicator

Results

Data table

A total of 9 students were enrolled in the program in Fall 2024, including 5 who were pursuing the degree as a co-terminal student and 4 as a traditional master's student. Among these 9 students, 6 took ENVE 506 in Fall 2024 and are thus used for program assessment in this cycle.

The following artifacts were used to assess performance toward each LO:

- LO #1 (Understand contaminant fate and transport in the environment):
 - HW assignments #4 and #7 and the final exam
- LO #2 (Formulate mass and energy balances and apply modeling concepts to practical environmental science and engineering problems):
 - HW assignments #1, #2, #3, #5, and #6
- LO #3 (Assess the sustainability of engineered systems):
 - Final report

Table 1 shows results from the 6 students in the program across LOs #1, #2, and #3. Scores range from 0 to 100 on a fractional basis. Definitions of proficiency levels are as follows: ≥ 0.9 (proficiency), 0.7-0.9 (meets expectations), < 0.7 (does not meet expectations). Distributions of proficiency levels are shown in Figure 1. Artifact samples are archived in Canvas.

Table 1. Results from evaluation rubric applied to LOs #1, #2, and #3 for this academic year

student	LO1	LO1 - DMP	LO2	LO2 - DMP	LO3	LO3 - DMP
student 1	0.82	M	0.88	M	0.91	P
student 2	0.55	D	0.93	P	0.91	P
student 3	0.84	M	0.76	M	0.81	M
student 4	0.90	P	0.76	M	0.81	M
student 5	0.84	M	0.95	P	0.91	P
student 6	0.83	M	0.73	M	0.81	M

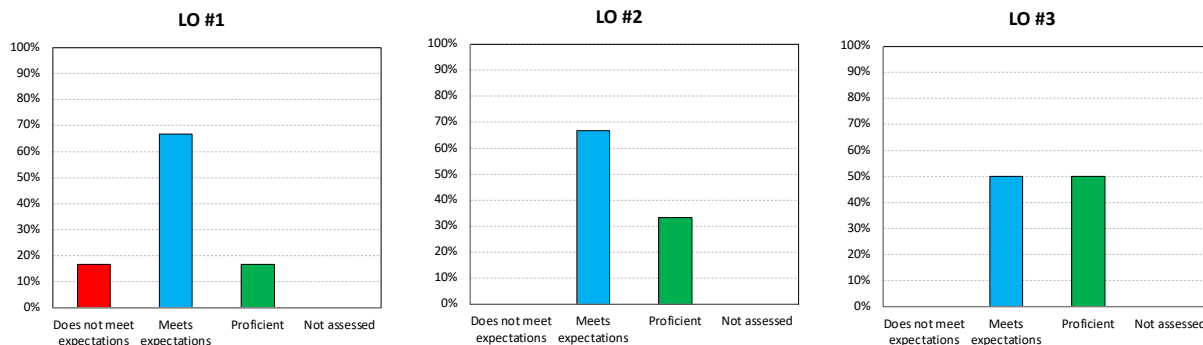


Figure 1. Distribution of proficiency levels for LOs #1, #2, and #3 for this academic year

Description of findings

Key findings from this academic year and cohort include:

- Five out of 6 students sampled met or exceeded expectations for LO #1, with one lone student below expectations primarily due to poor final exam performance (likely attributable more to personal matters than learning progress given the student's performance on other deliverables). Only 1 student exceeded expectations for LO #1, suggesting that on average students were reasonably prepared but not exceedingly prepared for the demands of the course and the program.
- All 6 students sampled met or exceeded expectations for LO #2 and LO #3, with a higher level of proficiency demonstrated overall.

Prior CI plan impact

In the previous assessment report, institutional reviewers made several recommendations to improve upon our assessment plan, including converting to an updated format, indicating more clearly which artifacts are used and collected, and to improve upon our CI plan. We had also previously noted that we must continue to keep a close eye on students coming from non-engineering backgrounds. In fact, advisors have attempted to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission and we continue to execute these continuous improvement actions. None of the performance shown above raises concerns for background preparation for the program.

This cycle's CI plan

1. In this cycle, we revised our learning objectives down from 5 to 3, crafting language that allows for a broader characterization of 3 main goals of the program. We also articulated a clearer plan for assessing the program using more than one course if needed, aiming to assess via two of the more frequently offered courses in the program that are also listed in the required courses list for the program (expanding from ENVE 506 only to include ENVE 506, ENVE 522, or ENVE 542). In this cycle, we had offered ENVE 506 with enrollment from a representative cross-section of the program, so we again used ENVE 506.
2. Continuing with last year's plan, advisors will attempt to improve preparation by students from non-engineering backgrounds by incorporating stricter prerequisite requirements at admission. We are paying closer attention to transcripts in the application review process, including scrutinizing what "engineering" means at different universities across the globe (i.e., in many countries architectural engineering lacks sufficient mathematics or physics backgrounds). In these cases, we now accept with deficiency course requirements of CAE 208 Thermodynamics and/or CAE 302 Fluid Mechanics, plus any required engineering mathematics courses that are missing.
 - a. None of the students in this cohort needed these prerequisite deficiency courses so this point is irrelevant for these students.